A multiplication rule for Motegi's type B/C trigonometric weight functions, in the fixed-point algebra

and a measured obstruction to a puzzle form

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Abstract

Motegi (arXiv:2610.15344, 11 October 2026) characterises a basis of K-theoretic factorial Schur P/Q-analogues — the type B and C trigonometric weight functions $\widetilde{WC}_{N,n,r}$, $\widetilde{WB}_{N,n,r}$ — by triangular vanishing, a diagonal evaluation that is a product over positive roots of C_N (resp. B_N), divisibility, smallness and uniqueness, and he never multiplies two of them. We supply the multiplication rule that those properties actually support. Let V be the algebra of functions on the 2^N specialisation points $\mathbf{u} = \mathbf{w}_{\mathbf{M}}$, $\mathbf{M} \subseteq [N]$, with pointwise product. We prove that the specialised weight functions form a basis of V (Lemma 5), that the resulting structure constants $c^{\nu}_{\lambda\mu}$ vanish unless $\nu \geq \lambda$ and $\nu \geq \mu$ (Theorem 8), that they satisfy a well-founded recursion (Theorem 9) whose extremal solutions are closed products over positive roots (Corollary 10), and that they admit a closed 2^N-term localisation formula built from Motegi's explicit matrix inverse (Theorem 11). We then record two measured negatives that bound what a combinatorial rule could look like: the identity does *not* lift to an identity between functions of $\mathbf{u}$ (Theorem 13), and the coefficients are *not* Laurent polynomials, with a sharp failure locus: measured over all 225 triples permitted by the support bound at $N = 2$ and $N = 3$, with no error in either direction, $c^{\nu}_{\lambda\mu}$ is a Laurent polynomial *if and only if* $|\nu| - \max(|\lambda|, |\mu|) \leq 1$, the smallest witness being $c^{(1,2)}_{\emptyset,\emptyset}$ at $N = 2$, which has a simple pole on the root hyperplane $w_1 w_2 = 1$ (Theorem 14). The second negative is an obstruction to any puzzle-style rule — but a *stratified* one: it bites only at cardinality jump ≥ 2, and it is not removed by rescaling the basis diagonally.

1 The census gate, and why this is not a puzzle paper

The session brief proposed building a type-A G_m puzzle with a reflecting edge and reading Motegi's structure constants off the configurations that touch that edge. A parameter census was run first, at position zero, before any code. Every cell below was read from a source file: Motegi's e-print source (`typebcweightfunctions_arxivversionfinal-3.tex`, 6430 lines, 216 486 B; line numbers below are into that file) and Wheeler–Zinn-Justin, *Littlewood–Richardson coefficients for Grothendieck polynomials from integrability* (the type-A G_m puzzle).

coordinate	Motegi WC/WB	type-A G_m puzzle (WZJ)	match?
alphabet variables	$\mathbf{u} = (u_1, \ldots, u_n)$, n; $x_i = 1 - u_i^{-1}$ (L552–561)	$x = (x_1, \ldots, x_k)$, k	count free both sides
equivariant / shift	$\mathbf{w} = (w_1, \ldots, w_N)$, N; $b_k = 1 - w_k$ (L561)	$y = (y_1, \ldots, y_n)$, n	one alphabet both
free deformation params	q **free** (stochastic six-vertex, L206–218); K-matrix β, γ (L333–343); Ikeda–Naruse K-theoretic β **fixed** $= -1$ (L527–530)	$q = 0$: the six-vertex R-matrix with the quantum parameter sent to zero (eq. (7)); $\beta = -1$ carried by the $-w$ weight of the K-tile	**NO:** 1 **vs** 0
closed under inversion?	**yes, as a coupling**: $(u_j u_k - 1)$, $(u_j u_k - q)$ (L478–480); crossings $R(u_i, u_j^{-1})$ (L397); $K(u)$ carries u^{-1}, u^{-2}	weights depend on $w = y_i/y_j$ only; products $y_i y_j$ never occur	differs; see (1)
arity of one weight	R: 2, through v/u. K: 1**, at an absolute position** (u^{-1}, u^{-2}, no ratio)	lozenge: 2, always the ratio y_i/y_j	**NO**
symmetry group(s)	**two**: S_n on $\mathbf{u}$ (rank $n-1$, sgn $= 0$, L481) *and* $W(X_N)$ on $w_j = e^{t_j}$, **rank** N, generators $s_0, \ldots, s_{N-1}$ (L858–872)	**two**: S_k on x and S_n on y (rank $n-1$)	first matches, **second does not**

Verdict. The gate did not kill the route; it corrected the brief on three counts, and the third redirected the session.

(1) *The pre-registered falsifier is unusable as stated.* The brief pre-registered "a type-A puzzle produces no $u_j u_k$ coupling at all", hence "a candidate rule producing type B/C functions with no boundary piece refutes either the rule or the formal-group connection". The coupling claim is confirmed on both sides, but the inference does not follow: Motegi's crossings are literally $R(u_i, u_j^{-1})$ (L397), and R depends on its arguments only through their ratio $u_j^{-1}/u_i = (u_i u_j)^{-1}$. So a type-$A$ R-matrix whose second alphabet is the *inverse* of the first already produces the coupling, with no new R-matrix. The true implication is the weaker "coupling $\Rightarrow$ the two alphabets are inverse to one another". This control therefore cannot falsify anything, and no claim is made from it.

(2) *The row that separates the two objects is the third, not the fourth: one free deformation parameter against zero.* Both objects sit on the G_m row of the formal-group table — spectral parameters combine multiplicatively on both sides — but at *different points of the quantum parameter inside that row.* The object named in the question, K-theoretic P/Q = Ikeda–Naruse $GQ_\lambda(x|b)$, $GB_\lambda(x|b)$, is Motegi's $q = 0$ specialisation (L552–561), whereas his characterisation theorems are for q free. "Read at G_m" is ambiguous between the two, and the puzzle side exists only at $q = 0$.

(3) *The brief's table has one row for two groups.* There are two symmetry groups on each side, over two different alphabets. The brief's reading names only the one that does the symmetrising (S_n on $\mathbf{u}$, type A, sgn $= 0$) — and on that row the two objects *agree*. The group that carries the type-B/C-ness is $W(X_N)$ of rank N acting on the $\mathbf{w}$-alphabet; it is the group whose inversion and ascent sets write the diagonal evaluation as a product over positive roots (L1000–1030, L1078). On the type-A side the corresponding group is S_n of rank $n-1$. Matched on the compared coordinate; separated on the uncompared one.

Consequence: (2) says the WZJ $q = 0$ puzzle will not multiply Motegi's q-free functions, and (3) says the structure actually available is the one indexed by $W(X_N)$ on the $\mathbf{w}$-alphabet — triangular vanishing in the order $\geq$ on subsets of $[N]$, a product-over-positive-roots diagonal, and an explicit inverse of the full specialisation matrix. That is an interpolation instrument, not a puzzle instrument. So no puzzle is built here. What is proved below is the rule those three facts support, together with a precise statement of what still blocks a puzzle.

2 Setup

Fix $N \geq 1$ and $X \in \{B, C\}$. Write $[N] = \{1, \ldots, N\}$ and $A^{\mathrm{c}} = [N] \setminus A$. Subsets are increasing tuples. For $A = \{a_1 < \cdots < a_s\}$ with $s \leq n$ put $\mathbf{w}_A = (w_{a_1}, \ldots, w_{a_s}, \underbrace{1, \ldots, 1}_{n-s})$, let $\lambda(A) = (a_s > \cdots > a_1)$ be the corresponding strict partition and $|\lambda(A)| = \sum_{a \in A} a$.

Definition 1 (Motegi's order, L640–650)**.** For $\mathbf{M}, \mathbf{I} \subseteq [N]$ with $s = |\mathbf{M}|$, $r = |\mathbf{I}|$, write $\mathbf{M} \geq \mathbf{I}$ iff $s \geq r$ and $m_{s-r+j} \geq i_j$ for $1 \leq j \leq r$. Equivalently $\lambda(\mathbf{I}) \subseteq \lambda(\mathbf{M})$ (L3248–3260).

Definition 2. Let $K = \mathbb{C}(q)(w_1, \ldots, w_N)$. Take $n = N$ (Motegi's choice, L4250–4256; by the stability relation `eq:typeC-stability-s-ge-r`, L2822–2828, any $n \geq N$ gives the same specialised values). For $\mathbf{M}, \mathbf{I} \subseteq [N]$ set

$$\widetilde{A}_{\mathbf{M},\mathbf{I}} := \widetilde{WX}_{N,N,|\mathbf{I}|}(\mathbf{w}_{\mathbf{M}} \,|\, \mathbf{w} \,|\, \mathbf{I}) \in K, \qquad \Delta_{\mathbf{I}} := \widetilde{A}_{\mathbf{I},\mathbf{I}}, \qquad A_{\mathbf{M},\mathbf{I}} := \widetilde{A}_{\mathbf{M},\mathbf{I}}/\Delta_{\mathbf{M}}.$$

Four facts are imported from Motegi; each was read first-hand from the e-print source and is cited with its line range.

(M1) Triangularity, all cardinalities. $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ unless $\mathbf{M} \geq \mathbf{I}$. *(Thm `ffunctionproperties(i)`, L603–612, proved for $F_{N,n,r}$ with no cardinality restriction; WC, WB are specialisations of F and inherit it, L787–790; $\widetilde{WX} = WX \cdot (\mathcal{E}^X)^{-1}$ with $\mathcal{E}^X$ independent of $\mathbf{I}$, L1054–1056; restated at general cardinality as $\lambda(\mathbf{I}) \subseteq \lambda(\mathbf{M})$, L3248–3260.)*

(M2) Diagonal evaluation over positive roots. $\Delta_{\mathbf{I}} = \prod_{\alpha \in \mathrm{Inv}(\lambda(\mathbf{I}))}(1 - e^{-\alpha}) \prod_{\alpha \in T(\lambda(\mathbf{I}))}(1 - qe^{+\alpha})$, written out in $\mathbf{w}$ at L1017–1022 and L1041–1046. *(Thm `typeCmainproperties(i)`, L1066–1082; type B at L1374–1390.)* In particular $\Delta_{\mathbf{I}} \neq 0$ in K: it is a product of binomials $1 - w_a^{-1}w_b^{-1}$, $1 - w_b/w_a$, $1 - qw_aw_b$, $1 - qw_b/w_a$, none of which is zero in K.

(M3) Laurent polynomiality of each specialisation. $\widetilde{A}_{\mathbf{M},\mathbf{I}} \in \mathbb{C}(q)[w_1^{\pm 1}, \ldots, w_N^{\pm 1}]$. *(Prop. L3310–3325.)*

(M4) Explicit inverse of the all-cardinality matrix. $\big(A(\mathbf{w}, q)^{-1}\big)_{\mathbf{I},\mathbf{L}} = q^{|\lambda(\mathbf{I})| - |\lambda(\mathbf{L})|} A_{\mathbf{L}^{\mathrm{c}},\mathbf{I}^{\mathrm{c}}}(\mathbf{w}, q^{-1})$. *(Thm `thm:BC-all-cardinality-biorthogonality`, L4364–4400.)*

Definition 3 (the fixed-point algebra)**.** Let $V := K^{\mathcal{P}([N])}$, the K-algebra of K-valued functions on the 2^N subsets of $[N]$ under *pointwise* product. For $\mathbf{I} \subseteq [N]$ let $e_{\mathbf{I}} \in V$ be $e_{\mathbf{I}}(\mathbf{M}) := \widetilde{A}_{\mathbf{M},\mathbf{I}}$.

3 The rule

Lemma 4. *$\geq$ is a partial order on $\mathcal{P}([N])$.*

Proof. Reflexivity is $m_j \geq m_j$. Antisymmetry: $\mathbf{M} \geq \mathbf{I}$ and $\mathbf{I} \geq \mathbf{M}$ force $s \geq r$ and $r \geq s$, so $s = r$; then $m_j \geq i_j$ and $i_j \geq m_j$ for all j, so $\mathbf{M} = \mathbf{I}$. Transitivity: let $\mathbf{M} \geq \mathbf{K} \geq \mathbf{I}$ with $|\mathbf{M}| = s$, $|\mathbf{K}| = t$, $|\mathbf{I}| = r$, so $s \geq t \geq r$, $m_{s-t+j} \geq k_j$ for $1 \leq j \leq t$, and $k_{t-r+j} \geq i_j$ for $1 \leq j \leq r$. Fix $1 \leq j \leq r$ and put $j' = t - r + j$; then $1 \leq j' \leq t$, and $m_{s-r+j} = m_{s-t+j'} \geq k_{j'} = k_{t-r+j} \geq i_j$. □

Lemma 5. *$\{e_{\mathbf{I}}\}_{\mathbf{I} \subseteq [N]}$ is a K-basis of V.*

Proof. $\dim_K V = 2^N = \#\mathcal{P}([N])$, so it suffices that $\widetilde{A} = (\widetilde{A}_{\mathbf{M},\mathbf{I}})$ is invertible over K. By Lemma 4, $\geq$ is a partial order on the finite set $\mathcal{P}([N])$; choose a linear extension $\trianglelefteq$ (so $\mathbf{I} \leq \mathbf{M} \Rightarrow \mathbf{I} \trianglelefteq \mathbf{M}$) and order rows and columns by it. If $\mathbf{M} \triangleleft \mathbf{I}$ then $\mathbf{M} \not\geq \mathbf{I}$ — otherwise $\mathbf{I} \trianglelefteq \mathbf{M}$, contradicting the strictness — so $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ by (M1). Hence $\widetilde{A}$ is lower triangular with diagonal entries $\Delta_{\mathbf{I}}$, and $\det \widetilde{A} = \prod_{\mathbf{I}} \Delta_{\mathbf{I}} \neq 0$ by (M2). □

Remark 6 (independent corroboration). Invertibility also follows from (M4), which exhibits A^{-1} outright. The two arguments are independent: Lemma 5 uses only triangularity and nonvanishing of the diagonal, whereas (M4) is Motegi's biorthogonality, derived from the type-A RTV orthogonality relation.

Definition 7. By Lemma 5 there are, for each $\lambda, \mu \subseteq [N]$, *unique* $c^{\nu}_{\lambda\mu} \in K$ with $e_\lambda e_\mu = \sum_{\nu \subseteq [N]} c^{\nu}_{\lambda\mu} e_\nu$, i.e.

$$\widetilde{A}_{\mathbf{M},\lambda}\, \widetilde{A}_{\mathbf{M},\mu} = \sum_{\nu \subseteq [N]} c^{\nu}_{\lambda\mu}\, \widetilde{A}_{\mathbf{M},\nu} \qquad \text{for every } \mathbf{M} \subseteq [N]. \tag{1}$$

Theorem 8 (support bound)**.** *$c^{\nu}_{\lambda\mu} \neq 0$ implies $\nu \geq \lambda$ and $\nu \geq \mu$.*

Proof. $\mathcal{P}([N])$ is finite, so $\geq$ is well-founded; we induct on ν along it. Fix ν and assume the statement for every $\kappa < \nu$. Suppose $\nu \geq \lambda$ fails. Evaluate (1) at $\mathbf{M} = \nu$. The left-hand side is 0, because $\widetilde{A}_{\nu,\lambda} = 0$ by (M1). On the right, (M1) kills every term with $\nu \not\geq \kappa$, leaving $\sum_{\kappa \leq \nu} c^{\kappa}_{\lambda\mu} \widetilde{A}_{\nu,\kappa}$. Let $\kappa < \nu$ with $c^{\kappa}_{\lambda\mu} \neq 0$; the inductive hypothesis gives $\kappa \geq \lambda$, and $\nu \geq \kappa$, so $\nu \geq \lambda$ by transitivity (Lemma 4) — contradicting the supposition. So no such κ exists, and (1) at $\mathbf{M} = \nu$ collapses to $0 = c^{\nu}_{\lambda\mu} \Delta_\nu$. Since $\Delta_\nu \neq 0$ by (M2), $c^{\nu}_{\lambda\mu} = 0$. Interchanging the roles of λ and μ gives the second half. □

Theorem 9 (the rule, as a recursion)**.** *For every $\nu \subseteq [N]$,*

$$c^{\nu}_{\lambda\mu} = \Delta_\nu^{-1} \Big(\widetilde{A}_{\nu,\lambda} \widetilde{A}_{\nu,\mu} - \sum_{\kappa < \nu} c^{\kappa}_{\lambda\mu} \widetilde{A}_{\nu,\kappa} \Big),$$

and by Theorem 8 the sum may be restricted to the finite set $\{\kappa : \lambda \leq \kappa < \nu,\ \mu \leq \kappa\}$. The recursion is well-founded, so it determines every $c^{\nu}_{\lambda\mu}$ from Motegi's specialisation values alone.

Proof. Evaluate (1) at $\mathbf{M} = \nu$, drop the terms with $\nu \not\geq \kappa$ by (M1), and divide by $\Delta_\nu \neq 0$. □

Corollary 10 (extremal coefficients in closed form)**.** *Put $S(\lambda, \mu) := \{\kappa \subseteq [N] : \kappa \geq \lambda,\ \kappa \geq \mu\}$.*

1. *If ν is minimal in $S(\lambda, \mu)$ then $c^{\nu}_{\lambda\mu} = \widetilde{A}_{\nu,\lambda} \widetilde{A}_{\nu,\mu} / \Delta_\nu$.*
2. *If $\lambda \geq \mu$ then λ is the least element of $S(\lambda, \mu)$ and $c^{\lambda}_{\lambda\mu} = \widetilde{A}_{\lambda,\mu} = \widetilde{WX}_{N,N,|\mu|}(\mathbf{w}_\lambda \,|\, \mathbf{w} \,|\, \mu)$.*
3. *$c^{\lambda}_{\lambda\lambda} = \Delta_\lambda = \prod_{\alpha \in \mathrm{Inv}(\lambda)} (1 - e^{-\alpha}) \prod_{\alpha \in T(\lambda)} (1 - q e^{+\alpha})$.*

Proof. (1) By Theorem 8 every κ with $c^{\kappa}_{\lambda\mu} \neq 0$ lies in $S(\lambda,\mu)$; minimality of ν leaves no such $\kappa < \nu$, so the sum in Theorem 9 is empty. (2) If $\lambda \geq \mu$ then $\lambda \in S(\lambda,\mu)$, and every $\kappa \in S(\lambda,\mu)$ satisfies $\kappa \geq \lambda$, so λ is least, hence minimal; apply (1) and $\widetilde{A}_{\lambda,\lambda} = \Delta_\lambda$. (3) is (2) at $\mu = \lambda$, with (M2). □

Theorem 11 (closed localisation formula)**.** *For all $\lambda, \mu, \nu \subseteq [N]$,*

$$c^{\nu}_{\lambda\mu} = \sum_{\mathbf{M} \subseteq [N]} q^{|\lambda(\nu)| - |\lambda(\mathbf{M})|}\, A_{\mathbf{M}^c,\nu^c}(\mathbf{w}, q^{-1})\, \frac{\widetilde{A}_{\mathbf{M},\lambda}\widetilde{A}_{\mathbf{M},\mu}}{\Delta_{\mathbf{M}}},$$

a sum of 2^N terms, each factor of which Motegi gives explicitly.

Proof. Divide (1) by $\Delta_{\mathbf{M}}$ and set $v_{\mathbf{M}} := \widetilde{A}_{\mathbf{M},\lambda}\widetilde{A}_{\mathbf{M},\mu}/\Delta_{\mathbf{M}}$; the system reads $v = Ac$. By Lemma 5, A is invertible, so $c = A^{-1}v$; substitute (M4). □

Remark 12. Theorem 11 is a genuine closed formula but a soft one: it is the fixed-point (localisation) form of the rule, manifestly neither supported in the sense of Theorem 8 nor positive nor visibly polynomial. The structural content is in Theorems 8 and 9 and Corollary 10, which are proved from triangularity and the root-product diagonal and not from (M4).

4 Two measured negatives

Theorem 13 (no lift out of the fixed-point algebra)**.** *For $n \geq N$ write $g_{\mathbf{I}}(\mathbf{u}) := \widetilde{WX}_{N,n,|\mathbf{I}|}(\mathbf{u} \,|\, \mathbf{w} \,|\, \mathbf{I})$ as a function of $\mathbf{u} = (u_1, \ldots, u_n)$. Then in general $g_\lambda g_\mu \neq \sum_\nu c^{\nu}_{\lambda\mu} g_\nu$ as functions of $\mathbf{u}$. Measured: at $N = n = 2$, $X = C$, the identity fails at a single generic point for* all 16 *pairs (λ,μ) (§5).*

Proof. (1) constrains the two sides only at the 2^N points $\mathbf{u} = \mathbf{w}_{\mathbf{M}}$, and the measurement exhibits a point at which the two sides differ, for every pair. A degree count makes this unsurprising: by the symmetrisation formula (L474–487), $WX_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})$ is $\prod_k u_k^{-(N+n)}$ times a polynomial of degree $N+n$ in each u_j, so a product of two of them has roughly twice the u-degree spread of the span of the g_ν. □

Theorem 14 (the structure constants are not Laurent polynomials)**.** *There exist N, λ, μ, ν with $c^{\nu}_{\lambda\mu} \notin \mathbb{C}(q)[w_1^{\pm 1}, \ldots, w_N^{\pm 1}]$. Explicitly, for $N = 2$, $X = C$, $\lambda = \mu = \emptyset$, $\nu = (1,2)$,*

$$c^{(1,2)}_{\emptyset\emptyset} = \frac{-q\, w_1 w_2 (q-1)^2\, P(q, w_1, w_2)}{w_1 w_2 - 1},$$

with P a polynomial not divisible by $w_1 w_2 - 1$; the pole lies on the root hyperplane $e^{t_1+t_2} = 1$, i.e. on the root $t_1 + t_2 \in \Delta^+(C_2)$. Moreover this is not *a diagonal-normalisation artefact: replacing $e_{\mathbf{I}}$ by $e_{\mathbf{I}}/\Delta_{\mathbf{I}}$, which multiplies $c^{\nu}_{\lambda\mu}$ by $\Delta_\nu/(\Delta_\lambda\Delta_\mu)$, reduces the number of Laurent coefficients from* 29/30 *to* 4/30 *at $N = 2$.*

Proof. Computed symbolically from Theorem 9 with the engine of §5, whose controls are reported there; the stated denominator is the exact output of `cancel` over $\mathbb{C}(q)(w_1, w_2)$. Note that Theorem 14 is consistent with (M3): each individual specialisation $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ *is* a Laurent polynomial, but Theorem 9 divides by Δ_ν, and that division need not be exact. □

Conjecture 15 (the failure locus, measured). $c^{\nu}_{\lambda\mu} \in \mathbb{C}(q)[w_1^{\pm 1}, \ldots, w_N^{\pm 1}]$ if and only if

$$|\nu| - \max(|\lambda|, |\mu|) \;\leq\; 1.$$

Measured for type C over every triple (λ, μ, ν) permitted by the support bound of Theorem 8: 30 triples at $N = 2$ and 195 at $N = 3$, 225 in all, with the criterion agreeing on every one and *no* error in either direction (no predicted-pole that was Laurent, no observed pole that the criterion missed). Laurent counts: 29/30 and 176/195.

Remark 16. The two numbers that make Conjecture 15 informative rather than curve-fitted are the *false-positive* count and the *false-negative* count, both zero: the criterion is not merely consistent with the failures, it also correctly predicts every success. It also explains the shape of the smallest witness: $\lambda = \mu = \emptyset$, $\nu = (1, 2)$ has jump 2, and it is the *only* failure at $N = 2$ because $N = 2$ admits no larger jump with λ or μ nonempty.

Remark 17 (what this rules out). A puzzle-style rule — a count of tilings with monomial weights — produces a Laurent polynomial. Theorem 14 therefore says that no such rule can compute $c^{\nu}_{\lambda\mu}$ for the basis in Motegi's normalisation. Any route to a combinatorial rule must first find a normalisation in which the coefficients *are* polynomial; Theorem 14 shows that the correction factor cannot be of the multiplicative form $g_\lambda g_\mu / g_\nu$ coming from the diagonal, since the obvious such choice makes matters worse. This is the brief's pre-registered fork resolved on the "wrong normalisation" side: it is *not* evidence against the G_m formal group.

5 Computational verification, with controls

All computation is pure Python + SymPy over $\mathbb{C}(q)(w_1, \ldots, w_N)$ (SageMath is not installed in this container). The engine (`scratch/q484_engine.py`) evaluates $F_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})$ by brute-force summation over S_n directly from the symmetrisation formula (L474–487), specialises to WC/WB (L494–505), divides by $\mathcal{E}^C_{N,n,r}$ (L1057–1062), and obtains $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ through the stability relation $\widetilde{WC}_{N,n,r}(\mathbf{w_M}|\mathbf{w}|\mathbf{I}) = \widetilde{WC}_{N,s,r}(w_{m_1}, \ldots, w_{m_s}|\mathbf{w}|\mathbf{I})$ for $s \geq r$ (L2822–2828), which removes the trailing 1's and so keeps the factors $(qu_j - u_k)/(u_j - u_k)$ nonsingular at generic $\mathbf{w}$.

Control 1 (diagonal, independent mechanism). $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ from the brute-force S_n sum, against the product over positive roots of $\mathrm{Inv}(\lambda(\mathbf{I}))$ and $T(\lambda(\mathbf{I}))$ computed from L1017–1022 and L1041–1046. 14/14 at $N = 1, 2, 3$, type C. The two sides are reached by different mechanisms (a sum over a symmetric group versus a product over a root system), so the agreement is corroboration, not a second reading of one engine.

Control 2 (triangularity). $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ if and only if $\mathbf{M} \not\geq \mathbf{I}$. 84/84 pairs at $N = 1, 2, 3$, with zero violations in *either* direction: 48 pairs with $\mathbf{M} \geq \mathbf{I}$ all nonzero, 36 pairs with $\mathbf{M} \not\geq \mathbf{I}$ all zero. (The "only if" direction is stronger than (M1), which asserts only vanishing.)

Control 2, lesioned. Replacing Motegi's order by the plausible wrong reading — componentwise $m_j \geq i_j$ with no cardinality shift — produces 1/16 violations at $N = 2$ and 8/64 at $N = 3$. The control fires.

Control 1, lesioned — and the domain fault it exposed. Two plants were made in the symmetrisation formula: P1 changes the reflecting-boundary pair factor $(u_j u_k - q) \mapsto (u_j u_k - 1)$; P2 drops the vanishing factor, $(u_j^2 - 1) \mapsto u_j^2$. P2 fires on Control 1 (3/4 labels at $N = 2$, 7/8 at $N = 3$). **P1 does not fire at all** (0/4, 0/8). The reason is a restricted domain: on the diagonal $\mathbf{M} = \mathbf{I}$ one has $s = r$, hence $n = r$, hence the index range $r + 1 \leq j < k \leq n$ is *empty* — three of

the seven product blocks of (L474–487), namely $(u_j u_k - q)$, $(\beta u_j - \gamma)$ and $\prod_k (u_j - q w_k)$ for $j > r$, are absent from Control 1's domain. Control 1 validates only the $j \le r$ part of the formula.

Control 3 (cross-sector branching), built to cover that domain. Motegi's cross-sector branching theorem (L2886) expresses the $s > r$ specialisation as an explicit sum over $\mathbf{K} \subseteq \mathbf{M}$, $|\mathbf{K}| = r$, of type-A weight functions $W_{N,r}(\mathbf{w_K}|\mathbf{w}|\mathbf{I})$; at $s = r$ the sum has one term (his Remark, L2924–2929), which lets $W_{N,r}$ be solved out of the engine's own $s = r$ values. The resulting identity has the full $n = s$ evaluation on the left (all seven blocks active) and only $n = r$ evaluations on the right (the $j > r$ blocks empty), so the planted factors occur on one side only. **Clean:** 27/27 (5 at $N = 2$, 22 at $N = 3$; $s > r$ pairs only). **Plant P1 fires**: 4/5 and 15/22. **Plant P2 is spared** (5/5, 22/22) — informatively, since P2 perturbs $j \le r$ factors, which appear on both sides of the branching identity and cancel. Controls 1 and 3 thus partition the formula between them, and each plant is caught by exactly one.

A control that did not fire and is reported as such. Motegi's all-cardinality biorthogonality (M4) was also implemented as a control. It is *clean at* 16/16 *($N = 2$) and* 4/4 *($N = 1$)*, but it is **insensitive to both plants** — P1 and P2 each pass 16/16 at $N = 2$ despite demonstrably altering matrix entries. It is therefore reported as a consistency check on the implementation of (M4), not as a check on the engine.

Main results, measured. Type C, over all pairs (λ, μ) and all labels ν. At $N = 2$ (16 pairs, 4 labels): Theorem 8 64/64; Corollary 10(2) 10/10 (every pair with $\lambda \ge \mu$); Theorem 11 64/64. At $N = 3$ (64 pairs, 8 labels): Theorem 8 512/512; Corollary 10(2) 35/35. (Theorem 11 was not re-run at $N = 3$: its 2^N-term sum over an 8×8 symbolic matrix did not fit the session.) Laurent polynomiality: 29/30 at $N = 2$ and 176/195 at $N = 3$, with the failures exactly the $1 + 19$ triples of Conjecture 15. Theorem 13: the lift to functions of $\mathbf{u}$ fails for all 16 pairs at the generic point $\mathbf{u} = (11/4, 7/5)$, $\mathbf{w} = (3/2, 5/3)$, $q = 2/7$.

6 Gaps and limits

The verdict of each item is printed beside the domain over which it was established.

1. **Proved, all N, both types, no computation.** Lemmas 4 and 5, Theorems 8 and 9, Corollary 10, Theorem 11. These rest only on (M1), (M2) and (M4), each of which is a cited theorem of Motegi read first-hand. No step uses $N \le 3$.

2. **Measured, type C only, $N \le 3$.** Theorems 13 and 14 are *existence* statements (one counterexample suffices), so the small domain does not weaken them. Theorem 13 rests on $N = n = 2$ alone. Conjecture 15 is a *universal* statement and is therefore the weakest item here: it is measured at $N = 2, 3$ only, and $N = 3$ is the first N admitting a jump of 3 — which it does not distinguish from a jump of 2, since both fail. So the criterion as stated (≤ 1 versus ≥ 2) is consistent with, but not evidence for, the absence of any further threshold. **A type B run was not performed at all**: every number in §5 is type C. Nothing in the proofs of Theorems 8–11 distinguishes the two types — they use only (M1), (M2), (M4), each of which Motegi states for both — but the negatives and the criterion are untested for B.

3. **Open: the combinatorial rule.** Q378's cell asks for a *rule*; Theorems 9 and 11 are a determination and a closed formula, not a combinatorial model. No puzzle, tableau or

increasing-tableau model is produced or conjectured here, and Theorem 14 shows that none can exist for this normalisation. The sharp open question this session produces is therefore: *in which normalisation of* $\{\widetilde{WX}_{N,n,r}\}$ *are the structure constants Laurent polynomials?* — with the partial answer that no diagonal rescaling achieves it.

4. **Not claimed: the geometric reading.** Motegi gives a localisation argument with $X_{\mathbf{M}} = \mathrm{Gr}(r, \bigoplus_j \mathbb{C}_{w_{m_j}})$ and T-fixed points (L3280–3330), which suggests that V is a localised equivariant K-theory ring and that the rule above is its multiplication table. I read that argument at chunk level only and do not assert the identification.

5. **Not claimed: the formal-group classification.** The organising principle that quantum integrable systems and generalised cohomology theories are both classified by one-dimensional formal groups is quoted from Zinn-Justin's lecture notes, *Lectures on Geometry, Quantum Integrability and Symmetric Functions*, p. 1 — a document marked "(IN PROGRESS)" of which I have read page 1 first-hand. It is used in §1 only to name the rows of a table, is not attributed to a theorem, and nothing below §1 depends on it.

6. **Trust grading of the imported facts.** In the registry (`proofs/registry/motegi-typeBC-multiplicatio`) the three imported facts are graded unequally, on purpose. (M1) is `proved`: its proof is Motegi's L620–790, which I read in full this session, and it carries both the triangularity *and* the explicit nonzero diagonal product `diagonalFfunction` — which is all that Lemmas 4–5, Theorem 8, Theorem 9 and Corollary 10(1)(2) require. (M2), the root-product *rewriting* of that same diagonal, has its proof deferred to Motegi's §3, which I did not read; it is graded `computed` on the strength of the 14/14 independent check, and it is needed only for Corollary 10(3). (M4) is graded `computed` for the same reason and is needed only for Theorem 11. So the main rule does not lean on anything graded below `proved`; the root-product form of the extremal coefficient and the localisation formula do.

7. **Retired control.** The brief's pre-registered row-4 falsifier (inversion coupling ⇒ a boundary piece is necessary) is withdrawn as unusable; see §1(1). No inference was drawn from it in either direction.